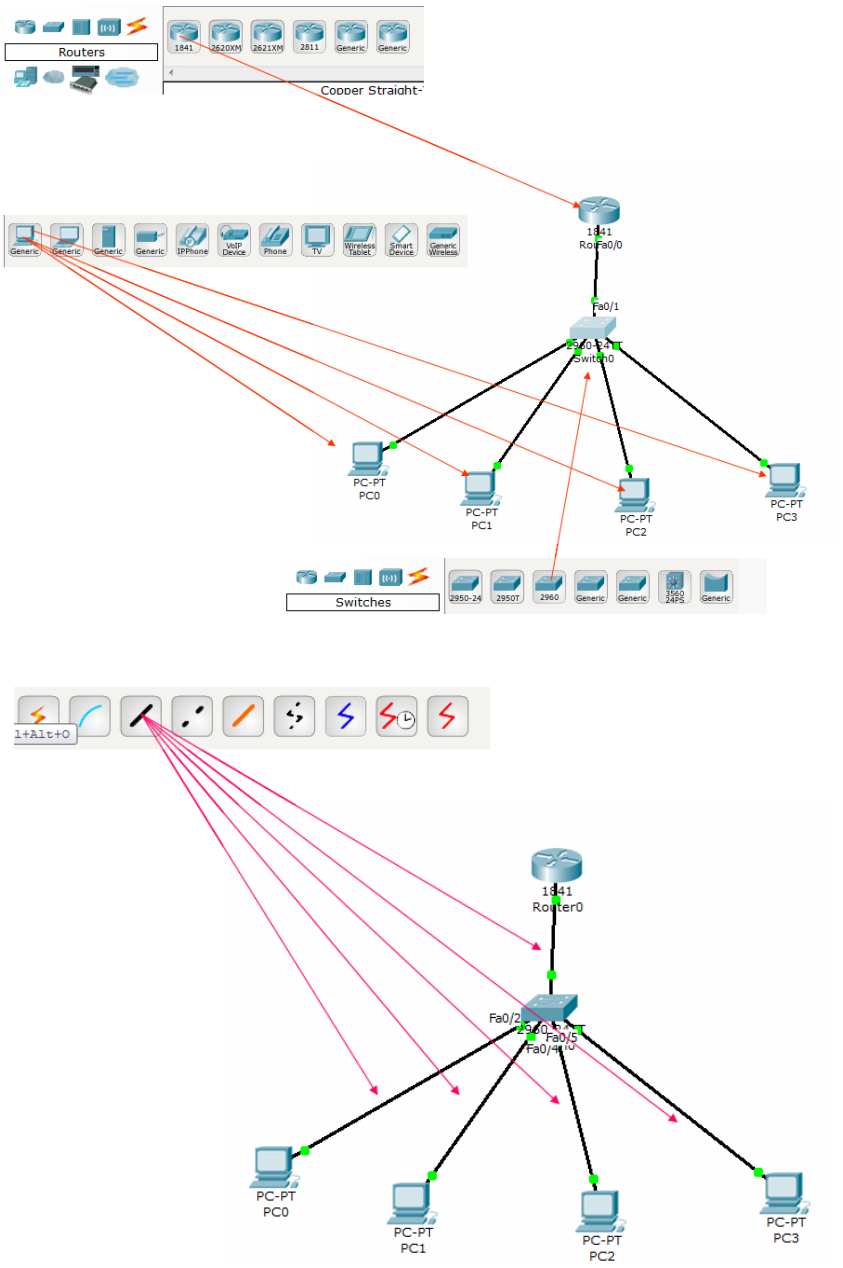
Expt-7
VLAN Configuration with Layer 3 Switch and Router

This experiment shows the way of configuring a VLAN switch under a router. During configuring a switch each interface is assigned under a VLAN (VLAN has both name and number) where the interface of a switch is referred to a port number. The router has to be configured such that each set of IP address (for a VLAN) is referred to a port number or interface of the router. Every port of the switch towards the PCs are in access mode where the port or interface of the switch towards the router is in trunking mode. During packet delivery, the router encapsulates a packet under IEEE 802.1q protocol where IEEE 802.1q header is added with each frame using the command 'encapsulation dot1q' and sends it to the appropriate port number.

Step-1
Implement the network like below.



The interface of devices are shown below.

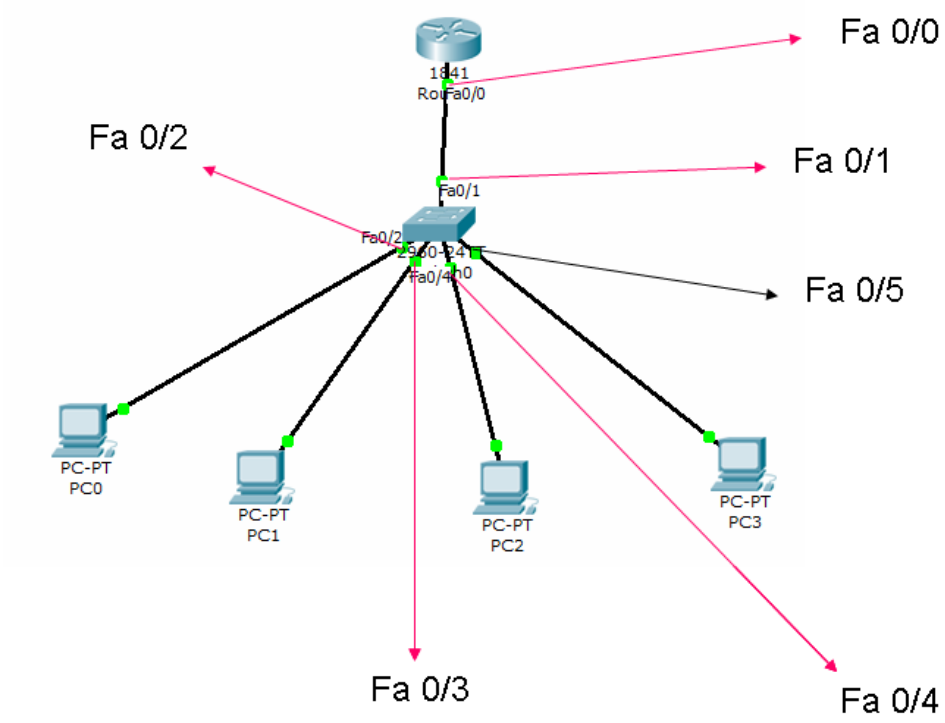


Fig.1(c)

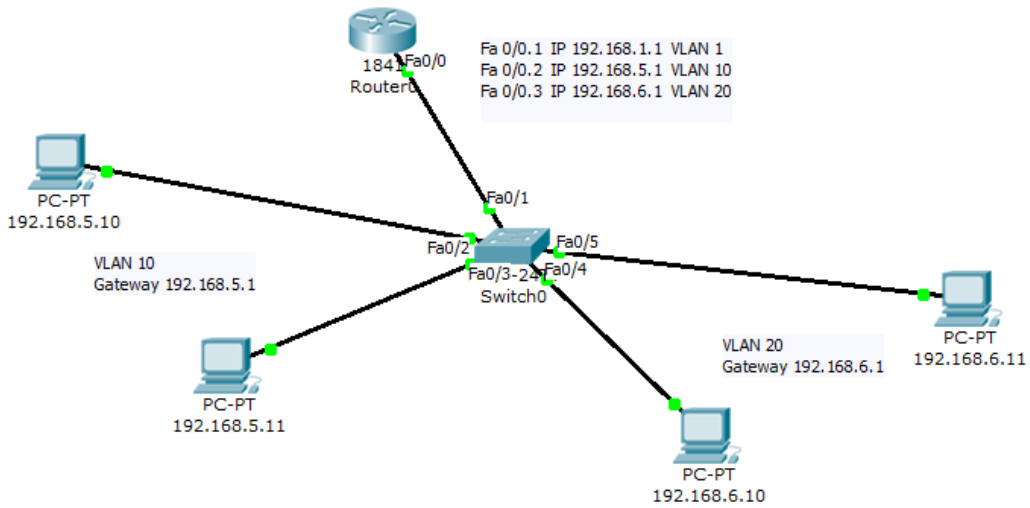
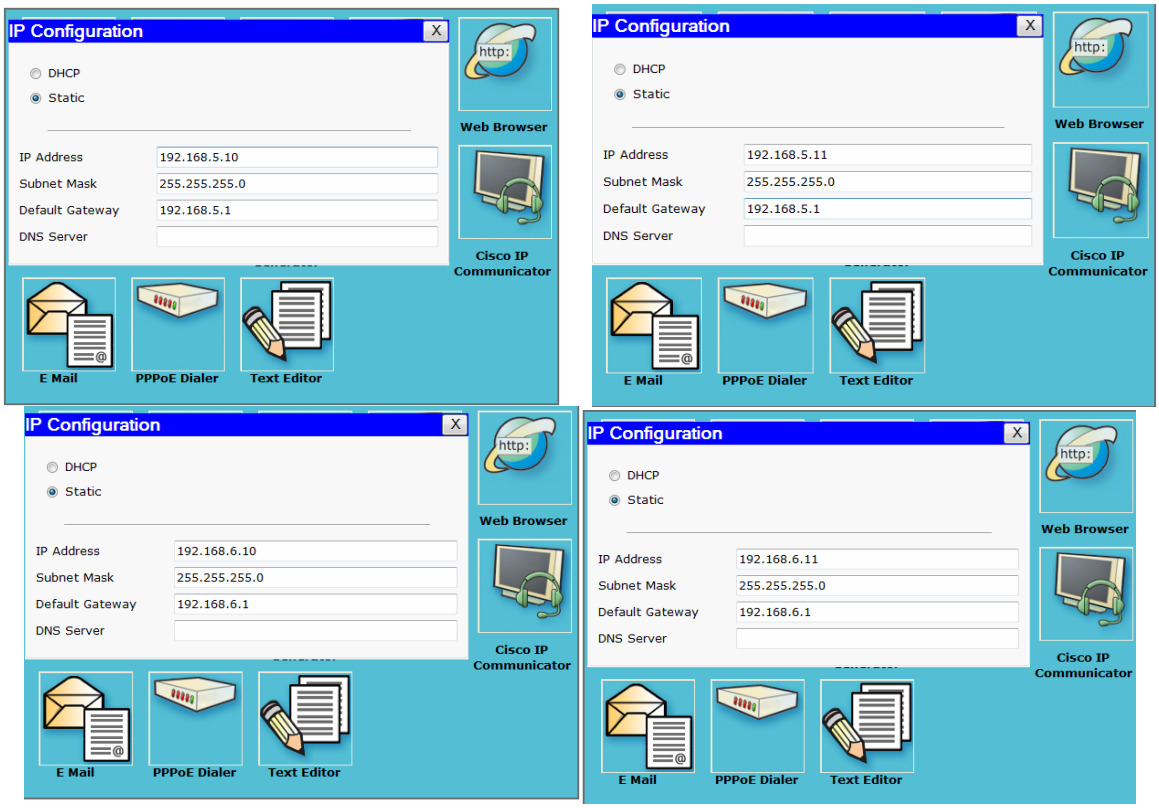


Fig. 1(d) Network Under VLAN 10 and 20

Step-2

Configure PC-0, PC-1, PC-2, PC-3 like below.



Step-3
Configure the switch like:

```
Switch>en
Switch#vlan database
% Warning: It is recommended to configure VLAN from config mode,
  as VLAN database mode is being deprecated. Please consult user
  documentation for configuring VTP/VLAN in config mode.
Switch(vlan)#vlan 10 name A
VLAN 10 added:
    Name: A
Switch(vlan)#vlan 20 name B
VLAN 20 added:
    Name: B
Switch(vlan)#exit
APPLY completed.
Exiting....
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int fa 0/2
Switch(config-if)#switchport mode access
```

```
Switch(config-if)#switchport access vlan 10
Switch(config-if)#int fa 0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#int fa 0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#int fa 0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int fa 0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#end
Switch#
```

Step-4

Configure the router like:

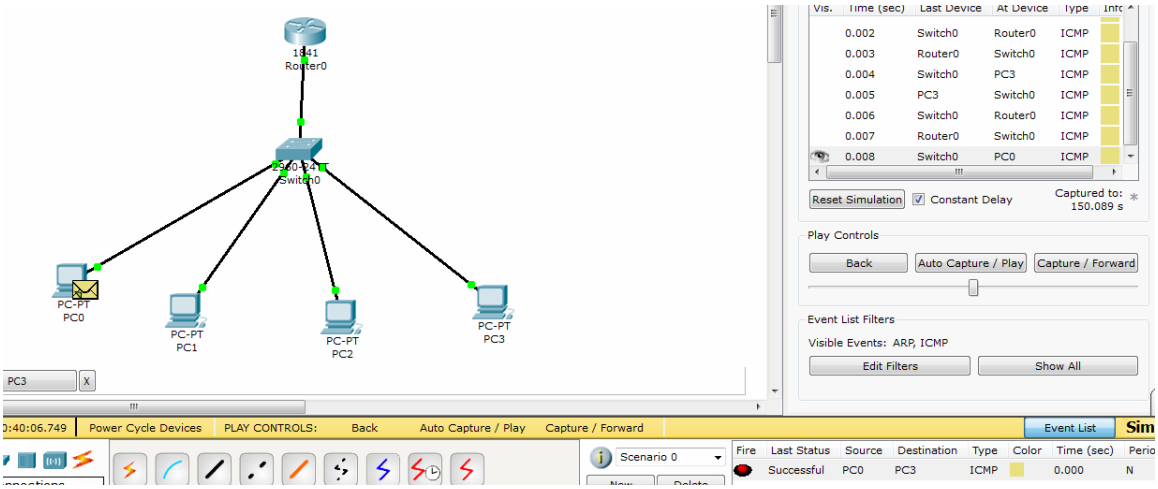
```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa 0/0
Router(config-if)#no shut
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config-if)#int fa 0/0.1
%LINK-5-CHANGED: Interface FastEthernet0/0.1, changed state to up
Router(config-subif)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.1, changed state to up
Router(config-subif)#encapsulation dot1q 1
Router(config-subif)#ip add 192.168.1.1 255.255.255.0
Router(config-subif)#int fa 0/0.2
Router(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.2, changed state to up
Router(config-subif)#encapsulation dot1q 10
Router(config-subif)#ip add 192.168.5.1 255.255.255.0
Router(config-subif)#int fa 0/0.3
Router(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.3, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.3, changed state to up
Router(config-subif)#encapsulation dot1q 20
Router(config-subif)#ip add 192.168.6.1 255.255.255.0
Router(config-subif)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
```

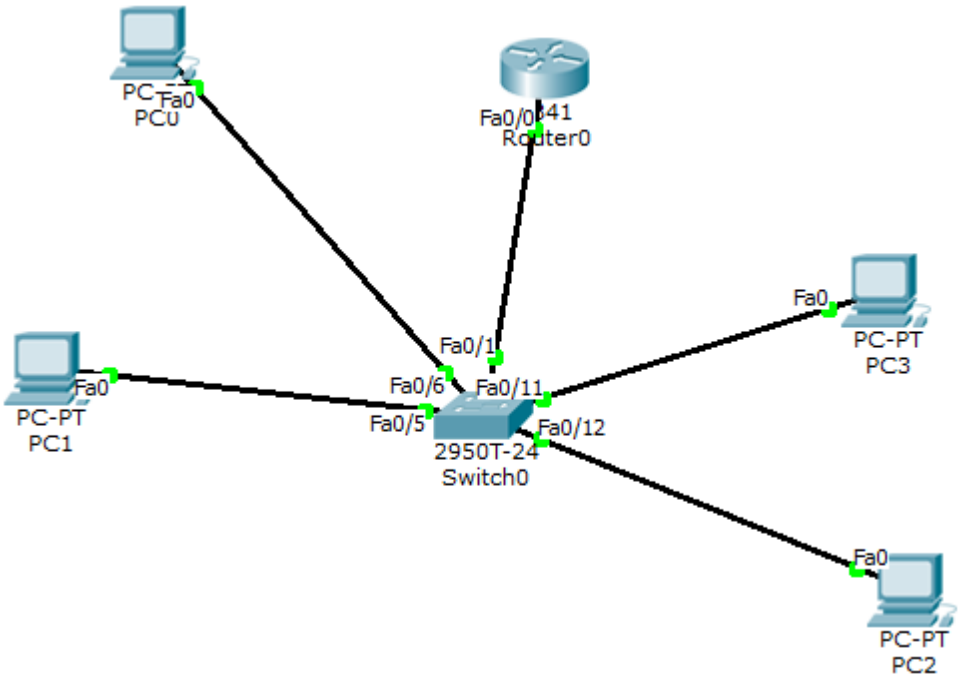
Step-5
In switch you can observe the VLAN configuration using following command.

```
Switch>sh vlan brief
VLAN Name                         Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                      active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                            Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                            Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                            Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                            Fa0/22, Fa0/23, Fa0/24, Gig1/1
                                            Gig1/2
10   A                           active    Fa0/2, Fa0/3
20   B                           active    Fa0/4, Fa0/5
1002 fddi-default                active
1003 token-ring-default              active
1004 fddinet-default             active
1005 trnet-default               active
Switch>
Switch>
```

Step-6
Verify the routing of packet under simulation mode. Packet transfer between any PCs within same VLAN (for example VLAN-10) will take place through the switch directly but between any two PCs of different LAN packet will be routed through the router.



We can implement above network using DHCP like below:



Step-1:
Let us configure the switch

Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.

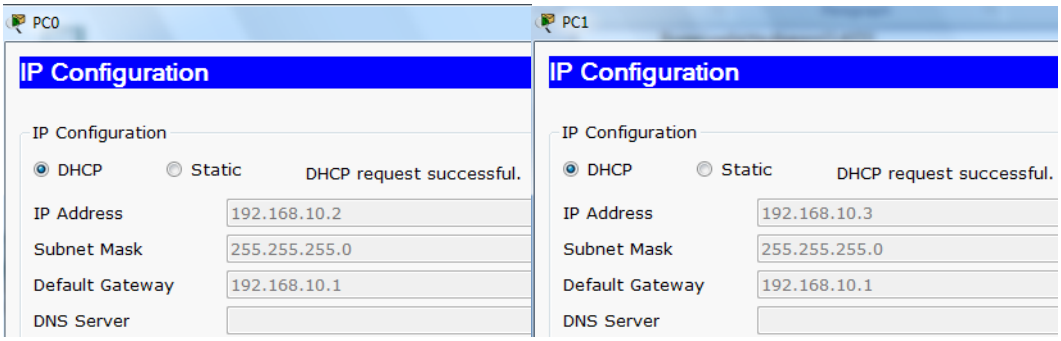
```
Switch(config)#int fa 0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#int fa 0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
% Access VLAN does not exist. Creating vlan 10
Switch(config-if)#int fa 0/6
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#int fa 0/11
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
% Access VLAN does not exist. Creating vlan 20
Switch(config-if)#int fa 0/12
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#end
Switch#
Router>en
```

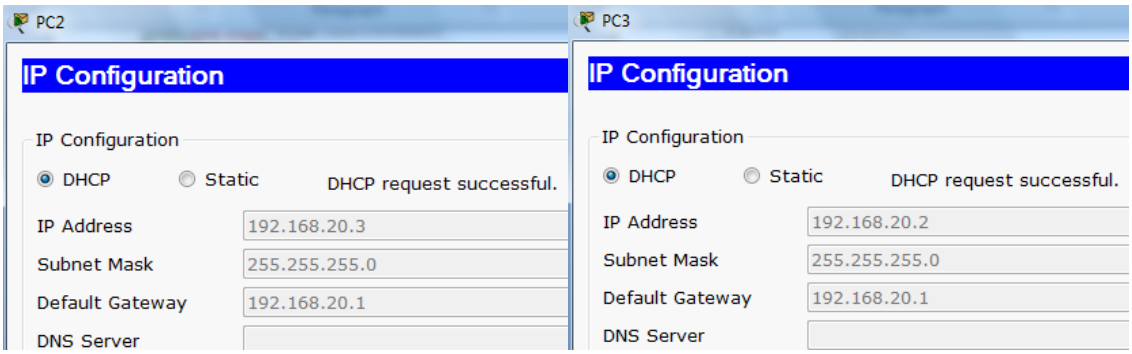
Step-2:
Let us configure the router

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa 0/0
Router(config-if)#no shut
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config-if)#int fa 0/0.10
Router(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.10, changed state to up
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip add 192.168.10.1 255.255.255.0
Router(config-subif)#int fa 0/0.20
Router(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.20, changed state to up
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip add 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#ip dhcp excluded-address 192.168.10.1  192.168.10.5
Router(config)#ip dhcp excluded-address 192.168.20.1  192.168.20.5
Router(config)#ip dhcp pool LAN10
```

```
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#exit
Router(config)#ip dhcp pool LAN20
Router(dhcp-config)#network 192.168.20.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.20.1
Router(dhcp-config)#end
Router#
```

Step-3
Click the PCs and select DHCP button then wait for selection of IP dynamically.





Step-4
Verify the flow of packet in simulation mode or use ping on command prompt.